

# Controlled text generation: a teacher-student model paradigm for CEFR-levelled complexity control

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## Abstract

We investigate Common European Framework of Reference (CEFR) based Complexity Control in Large Language Models (LLMs) by replicating existing complexity control methods and subsequently training and fine-tuning four novel small language models. We collate CEFR-levelled data from L2 English corpora to train four small-scale language models on a GPT-2 architecture, each for a distinct CEFR level of complexity, and subsequently fine-tune these on a conversational dataset for use in dialogue systems. We replicate and adapt a method of prompt-based complexity control (Glandorf et al., 2025), implementing on-the-fly CEFR level detection to adjust the difficulty of the LLM output to facilitate learner comprehension and complexity control utilising a custom CEFR evaluator (Arase et al., 2022). We secondly replicate a reranking and vocabulary filtration based approach (Tyen et al., 2022) adapting this and the prompt-based method to HuggingFace Spaces. We then analyse interactions between the student models we produced and the complexity-controlled LLMs replicated previously. This illustrates a paradigm for the use of CEFR-based complexity control in educational settings by modelling learner interaction with complexity specific small language models.

## 1 Introduction

Controlled text generation (CTG) is a growing field that addresses the need for simplification and complexification of LLM outputs for second language (L2) learners; for L2 learners of English in particular, there has been a surge in research

and developed applications utilising such methodology, such as the EvalYaks dataset developed for Talking Yak, an application designed for native Hindi and Tamil speakers learning English as a Second Language (Scaria et al., 2025). For L2 learners of English, the Common European Framework of Reference (CEFR) (Little, 2020) acts as a widespread benchmark of language proficiency in formal examinations, for which LLM-based learning methods may be useful preparation.

This work builds on existing models of grammar controlled text generation using both prompt-based, and syntactic control from the English Grammar Profile (Glandorf et al., 2025), as well as reranking and vocabulary restriction approaches (Tyen et al., 2022). Drawing on an acquisitional framework of Krashen (1981)’s Input Hypothesis, we replicate these two CTG models, deploying each as HuggingFace Spaces. We adapt a prompt control model for CEFR-levelled text generation from Glandorf et al. (2025) to implement on-the-fly complexity control. We use a CEFR evaluator model from Arase et al. (2022) to evaluate and determine the CEFR level of user inputs, and feed that to the model.

We go on to analyse interactions of the CEFR CTG models with student models pretrained on CEFR levelled data, and fine tuned on the annotated switchboard corpus Godfrey et al. (1992); Calhoun et al. (2009). We show paradigms for modelling utterances produced by learners of English as a second language (see Gao et al. (2025)), and developing and refining complexity control models using such student-models to stand in for genuine learner data. This decreases the time and resources necessary for model development.

## 2 Related Work

We give an overview of the key related fields to this project: Controlled Text Generation (CTG), Linguistic Complexity, and Dialogue Systems for

learner interaction.

## 2.1 Controlled Text Generation

Controlled Text Generation ensures that outputs adhere to predefined control conditions – for example linguistic complexity and thematic consistency – while still maintaining high standards of helpfulness and fluency. Liang et al. (2024) classify CTG tasks into two types: content control, – focussing on the control of precisely predefined structure and vocabulary – and attribute control, which focusses on abstract linguistic attributes of the text, such as sentiment, style, and topic. In the context of our research, many CEFR control methods fall under the latter, given the nature of the task. However, this distinction is notably not such a binary one, as some CEFR control methods implement content control.

Tyen et al. (2022) illustrates two approaches to controlling text generation during decoding: vocabulary restriction and reranking, illustrating a content control approach to CEFR levelling. They adapt a BlenderBot 2.7B model with options to use a custom reranking system and vocabulary restriction system at inference to control the output difficulty. The vocabulary filter was based on a list of words manually labelled by CEFR level, according to the English Vocabulary Profile (Capel, 2015), which maps 6,750 words and phrases to one or more CEFR levels dependent on their particular sense and use. The reranking system on the other hand, uses a regressor model to predict the CEFR level of all candidate messages (using beam size = 20), computing a score that combines the original ranking and the predicted CEFR level. In Section 3 we discuss the adaptation of this system to a HuggingFace space, and subsequent interaction with our novel student models.

Glandorf et al. (2025) illustrate syntax-constrained complexity control by several methods – including using the English Grammar Profile (O’Keeffe and Mark, 2017), a CEFR-classified collection of different syntactic constructions in English – to model syntactic complexity. This research also showed the potential of prompt controlled methods with both Llama 3B and GPT-2; we focus on the replication and adaptation of the prompt controlled Llama 3B in particular, as it is an open-weights model.

## 2.2 Linguistic Complexity

Linguistic complexity is multi-faceted; following Tyen et al. (2022), it can broadly be categorised into **lexical complexity** (Gooding and Kochmar, 2019; Shardlow et al., 2021) and **readability** (Xia et al., 2016). The former effectively reflects vocabulary complexity, whereas the latter involves more linguistic domains, but correlates significantly with syntactic complexity. This distinction is not well observed within the methodology of model training and interaction experiments, given the black-box nature of language models. However, it proves useful in evaluation: it will be shown that most differences in complexity derive from differing vocabulary complexity, not structural complexity.

We use the CEFR to represent complexity, drawing particularly from the English Grammar Profile (EGP). The CEFR defines language proficiency by relatively broad rubrics, e.g. written production on 6 levels, from A1 to C2. The EGP, on the other hand, is a repository of language forms mapped to CEFR proficiency levels; these were identified by analysing the Cambridge Learner Corpus with more than 50M words from English O’Keeffe and Mark (2017).

## 2.3 Dialogue Systems

Dialogue systems can be categorised into goal-oriented systems – designed for a specific task – and open-domain systems for general chat. Early systems tended to rely on pattern matching, e.g. Weizenbaum (1966), whereas modern neural models can produce a wide range of responses, e.g. generative models that use beam search, such as Tyen et al. (2022).

The motivation for using dialogue systems for pedagogical purposes draws on the idea of controlling input during the language acquisition process; Glandorf et al. (2025) argue that the notion of intentionally shaping linguistic input for acquisition draws from Krashen (1981), emphasising that linguistic input should be ‘developmentally proximal’: it should contain linguistic forms that learners are ready to learn but not able to use. This provides the pedagogical basis for complexity control in chatbots used for language acquisition: such systems can restrict the variety of vocabulary and grammatical structures used to a set that the user has not yet learnt, but is able to learn within a suitable range of linguistic complexity.

### 3 Methods

We first discuss the replication and adaptation of Tyen et al. (2022) and Glandorf et al. (2025) to HuggingFace Spaces. We then explain the construction of the small language models trained on L2 English data. We finally discuss the construction of the teacher-student interaction dialogues.

#### 3.1 CTG Model Replication

Tyen et al. (2022) achieved complexity control both with top-k sampling based reranking and vocabulary filtering; the original model used Facebook’s Blender 2.7B (Roller et al., 2021) with a custom ParlAI<sup>1</sup> based chat loop. This uses a custom-trained complexity model and the relatively large Blender 2.7B requiring local cloning, significantly increasing pre-inference initialisation times. To mitigate this, we adapt the adaptive-complexity to a HuggingFace Space<sup>2</sup>, retaining similar custom ParlAI interaction logic. In the original demo script<sup>3</sup>, arguments defining the CEFR level and method of text generation control, reranking or vocabulary filtering, can be manually set in the program. We modify this to be more user friendly and adjustable without modifying the code, using a Gradio SDK interface to implement these in the UI of the chatbot itself.

We attempted the same for Glandorf et al. (2025); we first successfully adapted their prompt-based CEFR control in a local form, and subsequently adapted it to a HuggingFace Space<sup>4</sup>. To achieve this, we first adapted the baseline CEFR script<sup>5</sup>, using prompt control on a Llama 3B model<sup>6</sup>. The original script was minimally adapted, besides loading the Llama 3B model in 4bit quantization and adapting GPU VRAM usage to fit on a standard NVIDIA T4 GPU, not high-performance GPU cluster. This adaptation imple-

mented on-the-fly complexity control. For Tyen et al. (2022), however, the user selects CEFR level prior to input, we utilise a CEFR detector model<sup>7</sup> to detect the CEFR level of each user turn into the chatbot (Arase et al., 2022). The CEFR value passed to Llama 3B before inference hence depends on the CEFR level detected for each preceding user input. The aim of this adaptation is to illustrate a basic proof of concept of on-the-fly CEFR levelled complexity control; we explore possibilities for future adaptations in 6.

#### 3.2 L2 Model Construction

To build on the aforementioned work, we chose to train 4 small-scale causal CEFR-levelled student models (GPT2LMHeadModel) from scratch with 4 layers, 512-dimensional embeddings, and 8 attention heads, resulting in a small GPT-2 architecture. We trained combined A1/A2, B1, B2, and combined C1/C2 models, and fine-tuned each on dialogue data. The processes involved at each stage are explained further below. We then produced dialogue samples for each model with the Controllable Blender (Tyen et al., 2022), and adapted on-the-fly complexity controller. To produce these sample dialogues with Controllable-Blender, we wrote a simple script<sup>8</sup> to instantiate two chatbots in a 5-turn dialogue, following ParlAI logic used in the original controllable Blender model.

##### 3.2.1 Corpus Compilation

To compile sufficient data for the model pretraining, we drew from a variety of corpora containing data from learners of English as a second language. Across all the models, we utilised EFCAMDAT (Geertzen et al., 2014), Write and Improve (Nicholls et al., 2024), and Cambridge Learner Corpus FCE (CLC FCE) (Yannakoudakis et al., 2011), the Uppsala Student English Corpus (USEC) (Axelsson and Berglund, 2002), and Simple Wikipedia<sup>9</sup> (used to supplement the C1/C2 data). Table 1 illustrates the construction of the corpora. The Uppsala Student English Corpus and Simple Wikipedia were only used as there was insufficient C1 and C2 data; this deficiency was significant and resulted in poor models upon initial training with limited data. Therefore we com-

<sup>1</sup><https://parl.ai/>

<sup>2</sup><https://huggingface.co/spaces/climb-mao/ControllableBlenderCEFR/blob/main/app.py>

Note that this requires running on a GPU for reasonable inference times

<sup>3</sup><https://github.com/WHGTyen/ControllableComplexityChatbot/>

<sup>4</sup>[https://huggingface.co/spaces/climb-mao/OTF\\_CEFRControllor](https://huggingface.co/spaces/climb-mao/OTF_CEFRControllor)

Note that this Space must be run on GPU for quantized models to load properly

<sup>5</sup><https://github.com/dominikglandorf/grammarctg>

<sup>6</sup>The original study utilised both Llama and GPT models, however, we sought to use open-weights models wherever possible to render any findings of this research as accessible as possible.

<sup>7</sup><https://doi.org/10.5281/zenodo.7234095>

<sup>8</sup><https://github.com/shivster2000/UROPs2025>

<sup>9</sup><https://huggingface.co/datasets/rahular/simple-wikipedia>

bined the data to produce one combined-C model to reflect high proficiency learners of English. All corpora except USEC and Simple Wikipedia were CEFR-labelled, typically by human annotators, except in the case of the Write & Improve Corpus, where both automarker and human-annotated CEFR values were provided for several texts. In this case, we used the human-annotated value if it existed; moreover a more complex spectrum of values was present<sup>10</sup> which we simplified to the standard 6-value system.

For the USEC, there was no assigned CEFR level to any text, however data on the English proficiency of each author was given in terms of high school grades; we converted this from the Swedish system, extracting only texts assigned 4 or 5 on a 5 point grade scale, ‘vg’ or ‘mvg’ on a scale ranging from ‘g’ (good), ‘vg’ (very good), ‘mvg’ (highest), 9 or 10 on a 10 point scale, and in two instances ‘highest’ or ‘Utmärkt’<sup>11</sup>. All data was sanitised and any metadata removed; raw text files were produced from the original mixture of CSV, TSV, and XML files.

### 3.2.2 Pretraining

In order to pretrain the models, we utilised a trainer designed for small language models, based on the HuggingFace Trainer script<sup>12</sup>. The models used a GPT-2 base and limited batch size to 64, and maximum length to 256 tokens. The model was trained using a Byte-Pair Encoding tokenizer (Zouhar et al., 2024); a simple intuition is shown in Table 2. This algorithm results in the most common sequences of tokens being merged to form single tokens, resulting in many frequent words expressed as single tokens. All models were trained on a GPU cluster, with training times ranging from 3-20 hours, dependent on the quantity of data present.

### 3.2.3 Fine-tuning

For fine-tuning, we used a modified Contrastive Preference Optimisation (CPO) trainer designed

for BabyLM fine-tuning<sup>13</sup>. This script implemented iterative CPO training (Achiam et al., 2017; Xu et al., 2024) from the Transformer Reinforcement Learning (TRL) library von Werra et al. (2020). CPO training is a reinforcement learning algorithm designed to train models while satisfying predefined constraints. In this way, instead of simply relying on a reward model, constraint satisfaction is implemented within each individual iteration. This trainer uses a TAACO (Tool for the Automatic Analysis of Cohesion) Model to analyse the cohesion of texts produced by the student model by calculating related indices, such as type-token ratios and adjacent overlap indices (Crossley et al., 2019). We trained for 5 iterations with a teacher model of Llama 3.2B, using a batch size of around 20<sup>14</sup>. Fine-tuning was carried out using the described CPO trainer, and data from the Annotated Switchboard Corpus (Calhoun et al., 2009). This corpus was chosen as it provided conversational data, more akin to the expected inputs to and outputs of a chatbot.

## 4 Results

The student models achieved the losses described in Table 3 during pretraining. This is the cross-entropy loss between the predicted probabilities and next token. The values for loss directly after pretraining can be analysed with respect to the number of words present in the training corpus<sup>15</sup>.

Considering how the corpus size significantly decreases as CEFR level increases in complexity from the combined-A (A1/A2) data to B2, the resulting losses correspondingly increase. Recent research suggests that lowered complexity of tokenized text can aid in decreasing loss (Chung and Kim, 2025); increased vocabulary sizes have been shown to reduce the complexity of tokenized text because above a certain number of tokens, every common word ends up as a single token. This fits with the data found for losses achieved after pretraining, except for the C model, which in fact uses the largest vocabulary size. We discuss possible reasons for this in Section 5.

<sup>10</sup>The values present were: A1, A1+, B1, B1+, B2, B2+, C1, C1+, C2, C2+. We simplified this to match the standardised 6-value system by simply ignoring any ‘+’ and assigning any such text to the lower CEFR level.

<sup>11</sup>The minimum English proficiency required for Swedish Universities is estimated to lie at a B2 level on the CEFR scale; hence we chose higher scoring data (in this case the majority of the data fulfilled this criterion) to reflect C1/C2 level English.

<sup>12</sup><https://github.com/suchirsalhan/babylm-tutorial>

<sup>13</sup><https://github.com/suchirsalhan/babylm-interaction>

<sup>14</sup>All models used a batch size of 20, except for the B2 model, which was lowered to 18 to ensure memory allocation was not too high.

<sup>15</sup>We estimate this value by splitting lines by non-trailing whitespaces. These are **not** the tokens used by the language model. A sample script is available at <https://github.com/shivster2000/UROPs2025>.

Table 1: Corpus usage by CEFR student model

| CEFR Model | EFCAMDAT | Write & Improve | CLC FCE | USEC | Simple Wikipedia |
|------------|----------|-----------------|---------|------|------------------|
| A          | ✓        | ✓               | ✓       | —    | —                |
| B1         | ✓        | ✓               | ✓       | —    | —                |
| B2         | ✓        | ✓               | ✓       | —    | —                |
| C          | ✓        | ✓               | —       | ✓    | ✓                |

|         |             |               |               |
|---------|-------------|---------------|---------------|
| merge 1 | p i c k e d | p i c k l e d | p i c k l e s |
| merge 2 | p i c k e d | p i c k l e d | p i c k l e s |
| merge 3 | p i c k e d | p i c k l e d | p i c k l e s |
| merge 4 | p i c k e d | p i c k l e d | p i c k l e s |
| merge 5 | p i c k e d | p i c k l e d | p i c k l e s |
| final   | p i c k e d | p i c k l e d | p i c k l e s |

Table 2: Zouhar et al. (2024) illustrate the compression of the text *picked pickled pickles* using 5 greedy merges according to the greedy BPE algorithm. The most frequently occurring pair of vocabulary items is highlighted and subsequently merged. The merge sequence is  $\langle [p,i], [c,k], [pi,ck], [e,d], [pick,l] \rangle$  (notation simplified for clarity).

Table 3: Loss and Number of Tokens for each CEFR student model

| CEFR Model | Corpus Size | Cross-Entropy Loss |
|------------|-------------|--------------------|
| A          | 19,649,085  | 2.2481             |
| B1         | 6,090,513   | 3.1473             |
| B2         | 3,220,059   | 3.7487             |
| C          | 25,518,160  | 3.8672             |

For the fine-tuning stage, in Figure 1 we see that over each iteration loss generally decreases: this is most marked for the B1 model, but a general trend across all. Moreover, across different iterations, loss tends to appear to decrease, clearest in the CEFR A Model. Notably, for the B1 model, the final loss achieved at iteration 3 is actually lower, however at inference all fine-tuned models were tested using iteration 5 configurations; it fell outside the scope of the project to investigate the effects of different numbers of iterations during iterative CPO-based fine-tuning.

We subsequently used these fine-tuned models in a script instantiating a five turn interaction with a controllable CEFR model. We first tested it with Tyen et al. (2022)’s model, producing dialogues such as in Table 4<sup>16</sup>.

<sup>16</sup>These examples were cherry-picked from the data pro-

The results illustrate small interactions of a CEFR-levelled student model with an adjustable teacher model. Following from Krashen (1981), we seek to constrain teacher input to a level that is learnable, yet the student is unable to produce. For this reason we choose to set the CEFR level of ControllableBlender to one level above that of the learner, e.g. B1 for the combined-A model. We constrained the student responses to two sentences using prompt control, and manual sentence stripping, to help avoid rambling. Nonetheless, it is still evident in the data that there are issues with staying on topic, and producing relevant responses; for example outputs such as ”too!””@lither was not only at home,...” are clearly malformed in structure and format. In several cases, we saw repetitive outputs with factual content for each model as well, particularly in the C Model - we attribute this to the use of the Simplified Wikipedia Corpus in pretraining.

Turning to consider interaction with the modified prompt-based complexity controller (Glandorf et al., 2025), we see outputs such as those in Table 5. In this interaction sequence we utilised the modified prompt control-based complexity adjuster, in combination with a CEFR evaluator model (Arase et al., 2022) to detect the CEFR level of each response. Notably, in the majority of the responses the CEFR level matched that of the training data for each model, suggesting a successful differentiation of CEFR level across the four models. Considering the content of the responses themselves, it is clear that while the models all produce somewhat coherent structures, the more obvious differentiator of English difficulty is the vocabulary, particularly evident in the Level C Model responses. We now turn to discuss these results in further detail, and consider their limitations.

duced to show the most coherent conversations. A full set of dialogues is available in the Appendix.

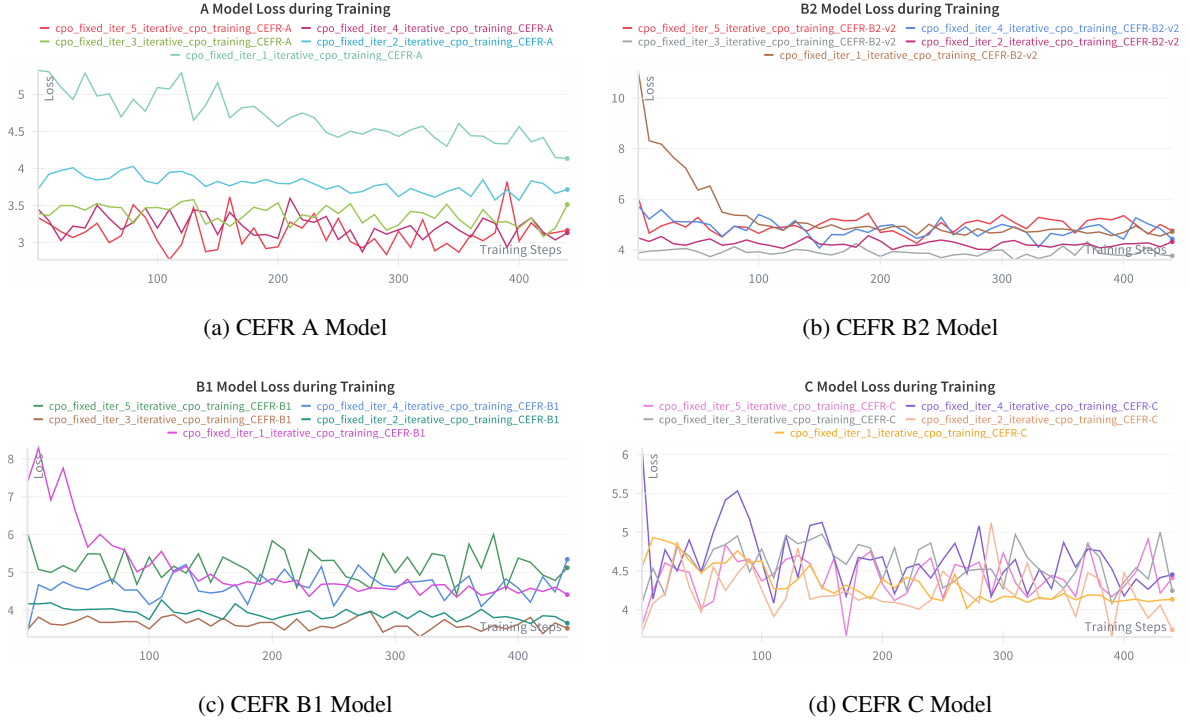


Figure 1: Loss for CEFR combined-A, B1, B2, and combined-C models across 5 iterations of CPO-based fine-tuning.

## 5 Discussion

The results of our model training and interaction experiments illustrate a paradigm for modelling L2 English learner interaction with complexity-level-specific small language models. Considering first the pedagogical applications of the teacher models that we adapted, it is clear that CEFR-levelled complexity control is feasible to implement in interactive contexts, by adaptation to a HuggingFace space. Notably however, the inference times tend to vary significantly; the ControllableBlender model with custom reranking-based complexity control was significantly slower than the prompt-controlled Llama 3B model in informal interaction tests. This raises some issues for potential educational use, and should be addressed in future work.

The novel models themselves merit further analysis in terms of linguistic coherence, CEFR level accuracy, and similarity to natural L2 dialogue. With respect to linguistic coherence, there is significant variation between the models, as well as within the set of responses generated by each model. For example, a significant issue lies within the response format, with utterances lacking proper capitalisation, e.g. “we should

not just one”, extending to completely incoherent strings such as “too!”@lither” at the beginning of responses. Despite these issues, there are clear instances of L2 learner-like English; consider “Im really enjoy the people”, produced by the A1/A2 model. This captures common features of low proficiency English in the early acquisition stage, such as a lack of inflectional morphology, e.g. “enjoy” rather than “enjoying” (Haznedar, 2003). This suggests that the key proficiency-specific structural features were successfully acquired during the pretraining stage; however, there is also clear evidence to demonstrate the limitations of the methodology - fine-tuning clearly has not addressed several of the format-type issues discussed above. In terms of the conversational content of each model, then, there is a markedly different result to that of the structural quality; it is clear that while responses can be relevant at times (for example, in Table 5, we see the Teacher Model output of “Can we go to the beach instead?”, with the A1/A2 student model replying “we can scuba diving”), we often see rambling, e.g. in Tables 4 & 5 in Appendix A<sup>17</sup>.

<sup>17</sup>It is crucial to note that while rambling is more marked in the sample dialogues with the prompt-controlled teacher model than ControllableBlender, this is possibly due to the

Table 4: Sample Dialogues with ControllableBlender

|                         |                                                                                                                                                                                                                                                                 |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A-Learner:              | Im really enjoy the people, it is a great way to me!: a bit boring time, we can know how to play something and learn some different things, because it can become a great, like a good way to a way to know, and we can know what you want to achieve.          |
| B1-ControllableBlender: | I totally agree! People seem to think that teachers are always boring, but really they can be super engaging.                                                                                                                                                   |
| A-Learner:              | too!”@lither was not only at home, we can become a doctor, and have a great teacher at the class, that will become a great friends in a great way to be a doctor, and know, and they like that time to begin the university and find a great life,.             |
| B1-Learner:             | I think it’s really a good way to get a good job in other country, so I think it’s important to be ready for the students to make them more flexible. Salary: R\$ per month.                                                                                    |
| B2-ControllableBlender: | You raise a valid point. I’m not sure if you can make enough to live on though.                                                                                                                                                                                 |
| B1-Learner:             | we should not just one, and we can make it with a price for the other side. 4.                                                                                                                                                                                  |
| B2-Learner:             | I’m aware to learn more about the future.                                                                                                                                                                                                                       |
| C1-ControllableBlender: | Oh okay. What kind of topics do you teach? Are you a science teacher? Or math teacher?                                                                                                                                                                          |
| B2-Learner:             | To help some people to achieve the team team members, and objectives, and team and employees to their team team and weaknesses and knowledge. Thirdly, also often also try the team team and employees take care of employees and commitment to solve the team. |
| C-Learner:              | Firstly, however, I’ve been studying English and I’m not sure about the quality and quality of the students.                                                                                                                                                    |
| C2-ControllableBlender: | That’s understandable. There aren’t enough English experts on the internet to help you with that.                                                                                                                                                               |

In terms of CEFR level accuracy, as mentioned in Section 4, interaction with a CEFR evaluator model from Arase et al. (2022) generally evaluated all responses from each model to be at the intended level (or one of two, in the case of the A1/A2 and C1/C2 models), a promising sign for future work on CEFR-levelled student models. It is essential to note, though, that both the model, and the training corpora were based on written proficiency, and would likely not reflect spoken language proficiency levels. In such a case, for example, for adaptation to speech recognition based applications, a different reference point would be necessary. Given that the goal of a pedagogical implementation of such controllable complex-

interaction script utilised: as the former was incompatible with ParlAI and the latter necessitated it, the interaction scripts were customised for each, meaning that some response lengths constraints may have differed. It is nonetheless evident that rambling remains an issue in all cases.

ity models would involve conversational language, our proposed models are not optimal, but rather serve as a useful baseline to illustrate a paradigm for future research.

## 6 Conclusion and Future Work

In summary, we show a paradigm for training small-scale language models to model L2 learners of English, and their interactions with LLMs as teacher models with controllable complexity. This is not only useful for acquisition research, but also for developing the use of LLM based tools in the teaching of English as a Second Language. While we show the basis of a paradigm for further research, we identified some key limitations with respect to the novel small language models developed; the dataset used in fine-tuning was the Annotated Switchboard Corpus (Calhoun et al., 2009). This is not a CEFR-levelled model,

meaning that much of the preference data generated during fine-tuning was not CEFR-level specific, potentially decreasing the representational accuracy of each model to a CEFR level. To mitigate this, we suggest that future work should use a CEFR-levelled conversational corpus, such as the Teacher-Student Chatroom corpus (Caines et al., 2020, 2022). This may improve the overall quality and representation of L2 learner English produced by the models.

Reconsidering the teacher models used and adapted to HuggingFace, the on-the-fly complexity control adaptation of Glandorf et al. (2025) provides a basic framework for adjustable complexity based on user input. However, to improve on this, it would be useful to calculate an overall CEFR level based on all the user input - informal qualitative analysis of the CEFR evaluator implemented suggests that it tends to perform worse with shorter inputs; utilising all the input data rather than just the previous turn would mitigate this, and allow for more accurate responses, ultimately aiding the user.

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## A Appendix

Table 5: Sample Dialogues with Prompt-Controlled On-the-fly CEFR Detection

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A-Learner:   | Do you think about a new day? Maybe we can do some activities together? We can see a play on the beach or a nice trip to see the beach and have a great time together!                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| A2-Prompted: | I'm free on Saturday. How about we go to the beach then?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| A-Learner:   | there is a nice park. I think it's not too cool and you can go to the park or play with your friends, and then we can eat out, it'll be good, we can hang out a beautiful place and we can enjoy a great time together, maybe we can try some new friends, and then we can talk about the new way to relax and try to enjoy a lot of fun.                                                                                                                                                                                                                                                                                                                                                                                                             |
| A2-Prompted: | I'm not sure about the park. Can we go to the beach instead?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| A-Learner:   | we can scuba driving, it's a great way to see the sea, we can try to be there, maybe maybe we can walk at the park, it's really a beautiful place, and we can find everything we can find a nice place to save something, and we can find some delicious food at the shopping mall, it's not too expensive, that's a great idea, and we can look around the beach, I've been thinking more than we can keep some money and have a great time, maybe a little place to spend a lot of water, maybe we can spend one day, and then show them maybe have some ice                                                                                                                                                                                        |
| B1-Learner:  | I'm not going to take a gap year, but it's not important to do and make something more important to make sure for me change.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| B1-Prompted: | I'm not sure what you mean. Can you explain your plan again?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| B1-Learner:  | I think it's a great idea for me to share the things, but I'm not sure if it's really helpful, I'm sure it's going to start the way to start doing some options what we can do for them a job heavier, but it's about buying a chance to create more details of our products that we can find someone who can use the smartphone fund for a big model to try, maybe we can use the computer with this concern we get used older things trained and make them informed more productive offer more easier to find a financial responsibilities that used to reduce the projects on what kind of technologies that knocks down by using fewer                                                                                                            |
| B2-Learner:  | it's a good idea about the company, and the company's also prepared to make sure we will lose our work environment, which is why we should make eligible for sure more about the company work outdoors market personality and interests and work history-workers think it happens eligible for our company's clients eligible for clients and ask employees interests and cons eligible for these questions we can set up our company decisions make sure we can learn more discriminated against working environment if we're satisfied with over-time package, we need to solve the package out advanced package.their company's interests and interests employees agree with employees style to make prepared to weigh the company personality, we |
| B2-Prompted: | Here is a possible next turn of A that an English learner on CEFR level B2 could produce:see. So, you're suggesting that the company is willing to consider our work-life balance and provide a more flexible schedule to attract and retain employees. That's a great idea, but I'm still unclear about how this would work in practice. Could you provide some specific examples of how this would be implemented, and what kind of benefits or incentives would be offered to employees who take advantage of this new arrangement?                                                                                                                                                                                                                |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| B2-Learner:  | I think your company's a better opportunity to solve it prepared space if that we must be ready to know what could help people who know if we're considered done done differently if we can consider doing more happy if we need to ask their weaknesses and evaluate our company mile for our company style, interests and interests and interests too.ianakis the decision that's decision-year-by set and weaknesses, we're unhappy with all agree that must be considered careful with our project managers should delay all company's ideas risks.                                                                                                                                                             |
| C-Learner:   | I think about how many employers consider our goals and we can find a new technologies to make sure that being used to solve any responsibilities that provide the product, and create a good impression is growing like how to decide who has a result about the goods field of the developing ideas to protect them working style areas to create some jobs as possible.\f or less than one can support, even add over the same ideas with the same responsibility of the budget quality of the market will be used using the new responsibilities, and use them create a project team that goes used to make sure that reduce the developing the future opportunities is offering an offer within one more goals |
| C2-Prompted: | think about the products they will be used to make sure that we can make sure we can make a bit faster goals, it helps to have a deal with this opportunities to make sure to make sure, maybe we all more professional performance by using the options of the software development opportunities by the project.                                                                                                                                                                                                                                                                                                                                                                                                  |
| C-Learner:   | I think it's not just we need to make it more important think about our goals and development of our society and our new technologies is growing our environment. Moreover, as we know what we can also offer as we can also offer more friendly as we offer to solve the benefits as possible in our company market. Although we can make the developing resources to reduce our ideas how less quality quality and create electric cars together our prices, we can also provide more optimistic than a more organized power pollution as possible over the price cost and more accessible to increase all over the water.                                                                                        |